React – Fundamentals

Session - 2

1. Create a new React app using Create React App, open App.js, and replace its content to display a JSX heading that says 'Welcome to React JSX!'.

Ans.

```
import './App.css';

// Functional Component
function UserGreeting(props) {
  return <h2>Hello, {props.username}!</h2>;
}

// Class Component
import React, { Component } from 'react';

class UserGreetingClass extends Component {
  render() {
    return <h2>Hello, {this.props.username}!</h2>;
  }
}

// Mini Profile Component
function MiniProfile() {
  return (
    <div className="profile">
      <img
        src="https://via.placeholder.com/150"
        alt="Profile"
        width="150"
        height="150"
      />
      <h3>Your Name</h3>
      <p>Learning React and building awesome apps! 🚀</p>
    </div>
  );
}

function App() {
  return (
    <div className="App">
      <h1>Welcome to React JSX!</h1>

      <UserGreeting username="Your Name" />
```

```
      <UserGreetingClass username="Your Name" />

      <MiniProfile />
    </div>
  );
}

export default App;
```

2. Write a functional component called UserGreeting that takes a username prop and displays 'Hello, {username}!' using JSX. Render it in App.js with your own name as the username.

Ans.

UserGreeting.js :

```
Create the project:import React from "react";

function UserGreeting({ username }) {
  return <h2>Hello, {username}!</h2>;
}

export default UserGreeting;
```

```
import React from "react";
import UserGreeting from "./UserGreeting";

function App() {
  return (
    <div>
      <UserGreeting username="John" />
    </div>
  );
}

export default App;
```

3. Convert the UserGreeting functional component into a class component named UserGreetingClass that also takes a username prop and displays the same greeting. Render both versions (functional and class) in App.js to compare.

Ans.

UserGreeting.js:

```
import React from "react";
```

```
function UserGreeting({ username }) {
  return <h2>Hello, {username}!</h2>;
}

export default UserGreeting;
```

```
import React from "react";
import UserGreeting from "./UserGreeting";
import UserGreetingClass from "./UserGreetingClass";

function App() {
  return (
    <div>
      <UserGreeting username="John" />
      <UserGreetingClass username="John" />
    </div>
  );
}

export default App;
```

4. Build a MiniProfile functional component that shows a user's display picture (use any image URL), name, and a short status message using JSX. Use this component to display your own mini profile like Instagram.

Ans.

MiniProfile.js:

```
import React from "react";

function MiniProfile() {
  return (
    <div>
      <img
        src="https://via.placeholder.com/150"
        alt="Profile"
        width="150"
      />
      <h2>John</h2>
      <p>Learning React 🚀</p>
    </div>
  );
}

export default MiniProfile;
```

App.js:

```
import React from "react";
import MiniProfile from "./MiniProfile";

function App() {
  return (
    <div>
      <MiniProfile />
    </div>
  );
}

export default App;
```

5. Refactor your MiniProfile component to ensure all JSX tags are properly closed and wrapped, and that no invalid HTML is used.

Hint: If you get any errors, check for missing tag closures or incorrect nesting in your JSX.

Ans.

MiniProfile.js:

```
import React from "react";

function MiniProfile() {
  return (
    <>
      <div className="profile">
        <img
          src="https://via.placeholder.com/150"
          alt="Profile"
          width="150"
        />
        <h2>John</h2>
        <p>Learning React 🚀</p>
      </div>
    </>
  );
}

export default MiniProfile;
```

App.js:

```
import React from "react";
import MiniProfile from "./MiniProfile";
```

```
function App() {
  return (
    <div>
      <MiniProfile />
    </div>
  );
}

export default App;
```